Final Project: the whole chain, in twelve minutes

Statistical Thinking for Data Science • First Semester, 2026/2027
Dr Abdulzeid Yen Anafo • University of Mines and Technology, Tarkwa

What this is

A **twelve minute presentation**, given in class, in which your group takes one real question from start to finish and shows how every stage of this course was used to answer it. Not a summary of the course. One question, carried the whole way, with the decisions you made and the ones you got wrong.
Groups of three or four, which you form yourselves. **Twelve minutes of talk, five minutes of questions.** Both are assessed. **The date will be announced in class and posted on the course page.**

The task

Choose a question that matters to you and that data can actually answer. Then do the work: find or collect the data, describe it, picture it, quantify the uncertainty, test or model what needs testing, and tell us plainly what your analysis does not support.

Your presentation must show the chain, in this order, because the order is the argument:

Stage	From	What we must see
The question	Week 2	the vague version you started with, and the answerable version you ended with
The data	Week 3	where the numbers came from, who is missing from them, and what that costs you
Description	Week 4	the distribution of your key variable, with its shape, not only a mean
Relationships	Week 5	at least one relationship, and the confounder you worried about
The picture	Week 6	one figure you are proud of, and one you rejected, with the reason
Uncertainty	Weeks 7 to 9	an interval on your headline number, and where it came from
A claim	Week 10	one pre-specified comparison or model, stated before you looked
A model	Week 11	a fitted relationship, read in the units of your problem
Honesty	Week 12	what your data cannot support, in specific terms

Where your data may come from

Any one of these:

1. **Data you collect yourselves.** A survey of a class, measurements in a laboratory, counts taken in the field, timings at the gate. Small is fine: forty honest observations beat four thousand downloaded ones you cannot explain.

2. **Public data.** Ghana Statistical Service, World Bank, Ghana Open Data, the Minerals Commission, a published supplementary dataset. Give the exact source and the date you downloaded it.

3. **The class wellbeing survey.** Permitted for this presentation only, because the presentation stays inside the course. It may not leave this room: no posters, no final-year projects, no publication, and no copies outside the course, as set out in Week 12.

Data about identifiable people needs their consent and must be anonymous in your slides. Never show a table cell containing one or two people.

What to submit

1. **Your slides**, as a PDF, at most twelve slides, uploaded before the session begins. The upload link and the deadline will be announced.
2. **Your data and your code**, as one zip or a link: the dataset as you used it, plus the script that produced every number and figure in the talk. A number nobody can reproduce does not count.
3. **A one page data memo**, which states the source of the data, how the sample was obtained, every change you made to the raw file, and the analysis plan you fixed before looking.

How it is marked

Out of 100. Nine tenths of the marks are for thinking, and a tenth for delivery.

Component	Marks
A question the data can answer, narrowed well and stated precisely	15
Provenance: where the data came from, who is missing, what that costs	15
Description and figures: honest, labelled, chosen for the comparison	15
Uncertainty: an interval on the headline number, correctly interpreted	15
Inference or model: pre-specified, correctly read, effect size reported	15
Limits: a specific, convincing account of what the data cannot support	15
Delivery: timing, clarity, and answers under questioning	10

Marks are shared by the group, except for the ten delivery marks and the questions, which are individual.

The limits section is worth as much as the analysis. A group that finds nothing and explains clearly why their design could never have found it will score above a group that reports a striking result they cannot defend. This is deliberate, and it is the whole point of the course.

The questions at the end

Every member of the group answers at least one question, and I choose who answers which. Expect questions of this kind:

- Why is your interval that wide, and what would make it narrower?
- Who is missing from your data, and which way would their absence push your answer?
- Show me the line of code that produced this number.
- You adjusted for programme. What would change if you had not?
- If I doubled your sample, which of your conclusions would change?
- What did you decide before you looked at the data, and what after?

Answering "my group mate did that part" scores zero for that question. Divide the labour however you like, but everyone must understand the whole chain.

Using AI tools

You may use them, and you must say so. Add a final slide listing every tool you used and what you used it for: cleaning data, writing code, drafting text, making figures. That slide costs you no marks. Its absence, or a tool used and not declared, is academic dishonesty and goes to the Department.

Be warned about one thing in particular. A language model will happily produce a confident paragraph of statistical interpretation that is subtly wrong, and the questions at the end are designed to find exactly that. If you cannot explain a sentence on your slide in your own words, take it off the slide.

Timetable

Step	What happens
Group and question	register your group and one paragraph question with me in class
Check-in	bring your data and your analysis plan, before you analyse anything
Upload	slides, data, code and memo, through the link announced in class
Presentations	in class, twelve minutes plus five of questions

Dates

All dates will be communicated in class and posted on the course page. Nothing here is due until you hear the date from me. Register your group early anyway: the questions that work well take a week of thinking, and the ones chosen the night before are visible from the back of the room.

Advice, from the marker

1. **Shrink the question.** "Does mining affect water quality in Ghana" cannot be answered by anyone in one semester. "Is turbidity higher downstream than upstream of one site, on the days we sampled" can.

2. **Spend your time on the data, not the model.** Most marks are lost at stage two, not stage eight.

3. **Write the limits slide first.** If you draft it in week one, it tells you what to fix while there is still time.

4. **Rehearse with a timer.** Twelve minutes is four or five slides of real content. Groups that overrun are stopped, and the limits slide is the one that never gets shown.

5. **Show one number you got wrong.** The groups that say "we first computed this, then realised the sample was selected" are the ones that convince me they did the work.